

PAX vibration diagnostics — run report

RUN

203850_pax-vibration_pax-vibration

Status: completed | Started: 2026-09-24T19:38:50.174193+00:00

PURPOSE

Compare fast PD variance with the PAX motor running and stopped. Log all fresh PAX telemetry and static sphere-angle variance during the on window. Duration is per condition; no actuator waveform.

REQUIRED OPTICAL SETUP

PD on selected IN1/IN2 with DC coupling and unchanged light/gain in both conditions. Keep PAX physically mounted and optics fixed. Motor-off means rotation stopped, not power unplugged. Outputs hold the configured static biases (default zero).

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

OPERATOR SETUP NOTES

V_{pi} expectation only; verify channel and measured driver gain before calibration.

DATA

54 recorded samples; pax_on: 26, pax_off: 28

Source: data.csv; full acquisition settings: recipe.json; status: run.json

PAX motor noise — variance, band [5.0, 1000.0] Hz

PAX_ON

sample_count: 425984
acquired_s: 27.917287423999994
wall_span_s: 30.729239230015082
sampling_rate_hz: 15258.789062499998
dark_corrected_mean_v: 0.008968105109838339
normalized_variance: 0.002881585736494456
normalized_band_variance: 0.0006819435872456251
status: ok

PAX_OFF

sample_count: 458752
acquired_s: 30.064771071999992
wall_span_s: 30.81515233698883
sampling_rate_hz: 15258.789062499998
dark_corrected_mean_v: 0.008635934176189559
normalized_variance: 0.003885790440311048
normalized_band_variance: 8.477439584499117e-05
status: ok

COMPARISON

normalized_variance_on_minus_off: -0.001004204703816592
normalized_variance_on_over_off: 0.7415700307975929
normalized_band_variance_on_minus_off: 0.000597169191400634
normalized_band_variance_on_over_off: 8.044216422285682
interpretation: Signed on-minus-off excess, not proof of mechanical causation. Compare the same PD bandwidth, optical alignment, light level and gain. Sequential drift, shot noise and electronics can also differ. PAX power/angles have a much slower bandwidth.

PAX ON: STATIC POLARIZATION / POWER

sample_count: 208
mean_power_w: 0.0005897880197115384

PAX motor noise — variance, band [5.0, 1000.0] Hz

normalized_power_variance: 0.004857003170729862

u_variance_rad2: 0.008869818893533976

v_variance_rad2: 0.006143177722965522

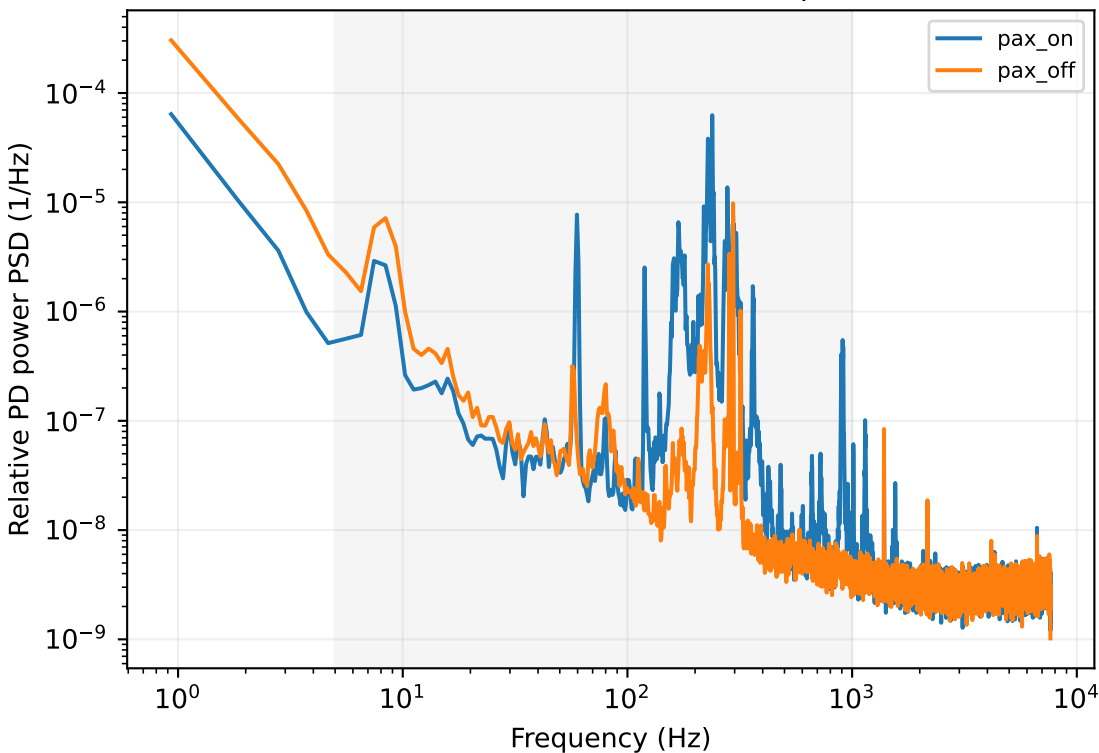
mean_dop: 0.9430139423076924

minimum_sin_v: 0.5837689598839996

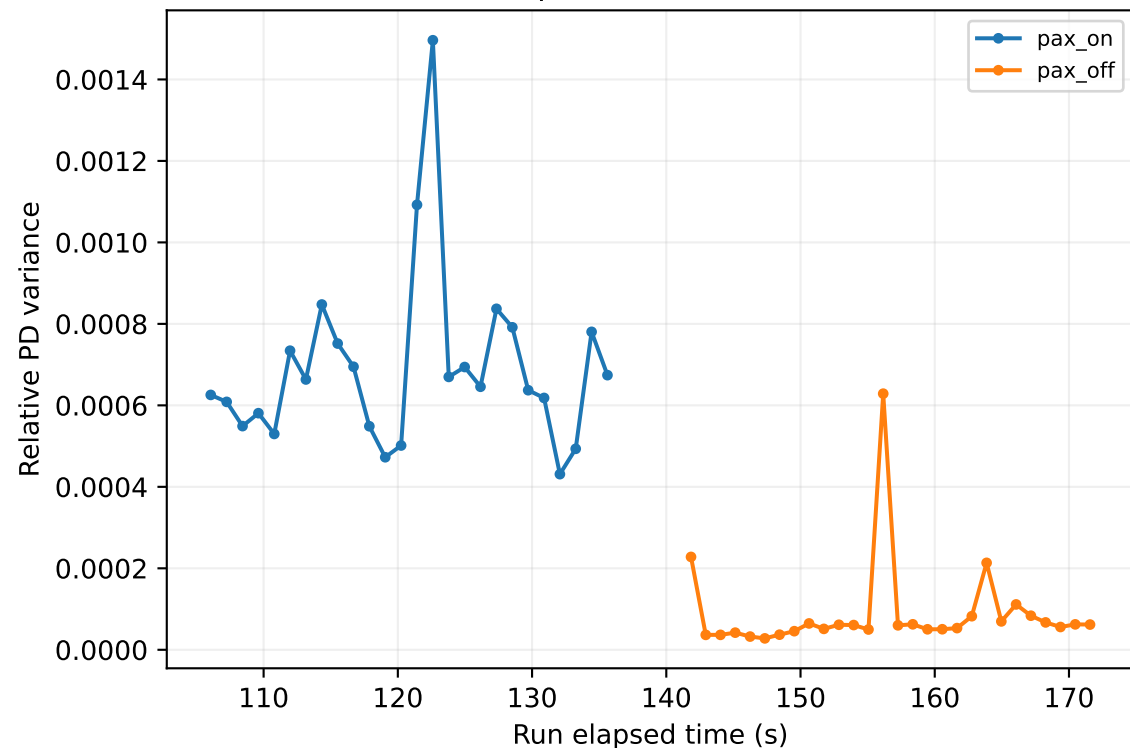
angle_note: u unwrapped in acquisition order; coordinate noise grows near S1 poles

update_rate_hz: 6.772456845669043

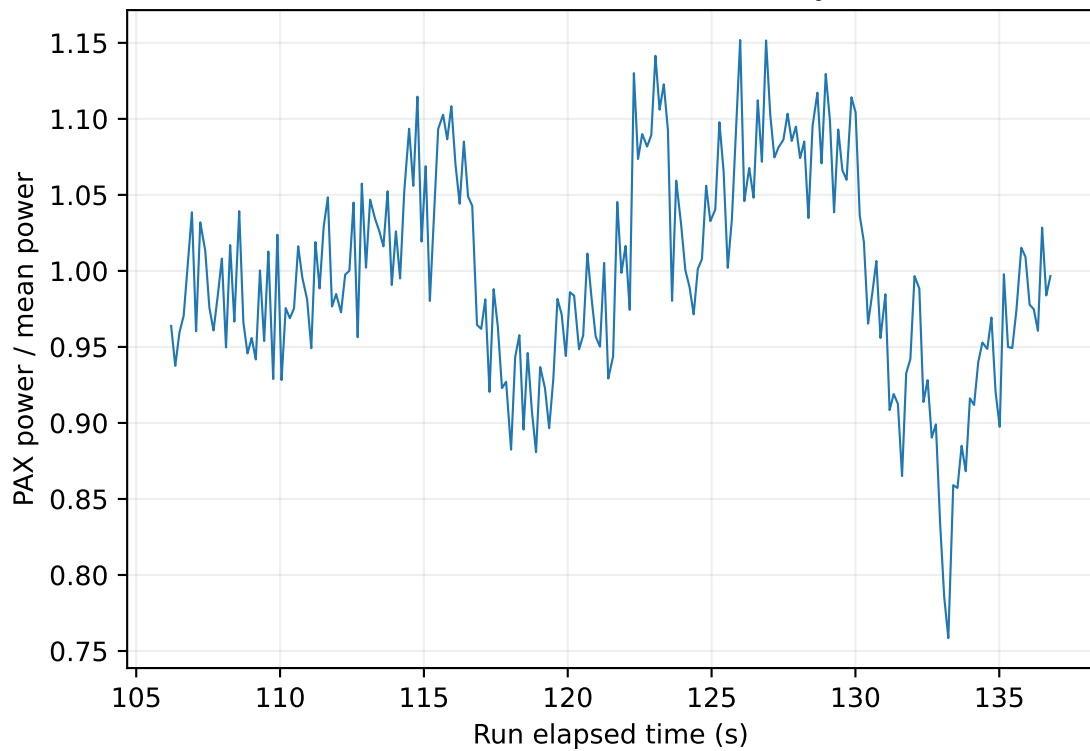
Identical PD bandwidth; shaded comparison band



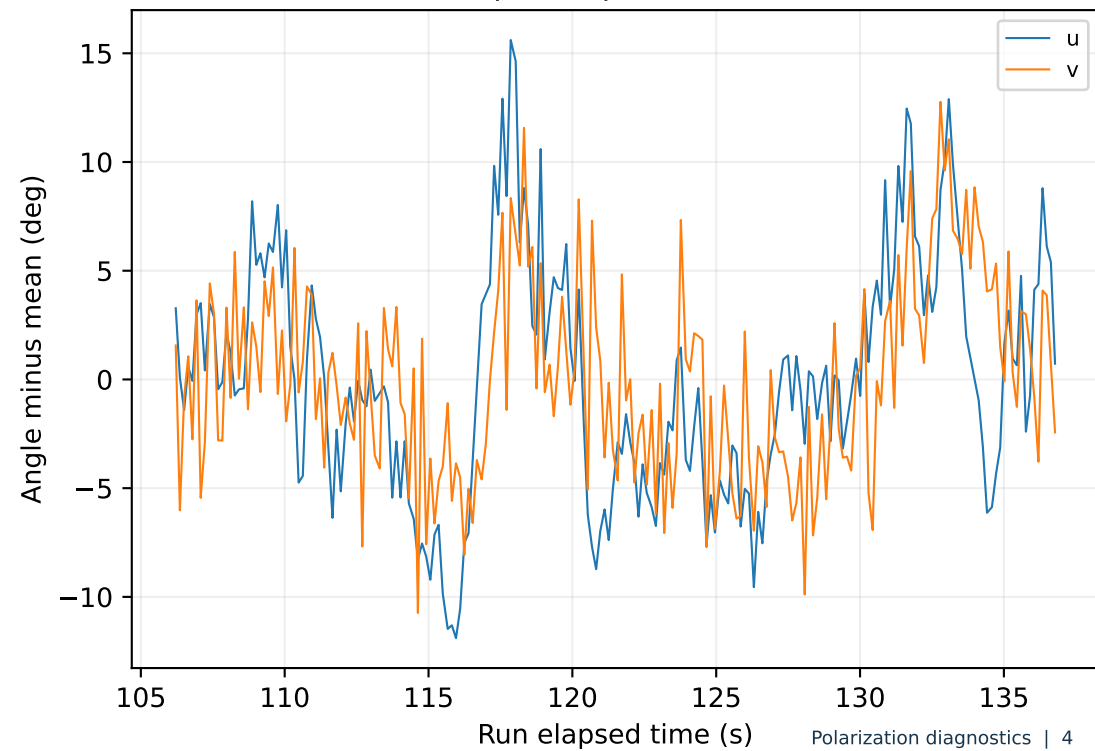
Per-capture band variance



PAX on: slow reference only



Static S1-polar sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"pax-vibration"
duration_s	30.0
report	"pdf"
pd_input	"in2"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
pax_vibration_scope_decimation	8192
pax_vibration_band_low_hz	5.0
pax_vibration_band_high_hz	1000.0
pax_vibration_motor_settle_s	5.0
pax_vibration_dark_voltage_v	null
pax_vibration_phi1_bias_voltage	0.0
pax_vibration_phi2_bias_voltage	0.0